Saramsha Dotel

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Experienced software engineer & technical leader with 7-year track record of planning, designing and building production-grade systems. I specialize in developing AI systems with strengths across backend, data engineering & cloud technologies and have led and scaled technical teams & projects.

KEY SKILLS

- **Languages:** Python, Javascript/Typescript
- **Backend Engineering:** System Design, RESTful APIs, GraphQL, Django, FastAPI, Flask, NodeJS
- **Databases:** PostgreSQL, SQL, DynamoDB, ElasticSearch, Redis, Vector Databases
- **AI & Machine Learning:** Large Language Models (LLMs), Agentic AI, Retrieval Augmented Generation (RAG), Harness Engineering, Prompt Engineering, Predictive Modeling, MLOps
- **Cloud & Platform Engineering:** Docker, Terraform, AWS (EC2, ECS, S3, Lambda, ECR, CDK, Cloudformation, Sagemaker), GCP, Grafana, Prometheus, CI/CD, DevOps Engineering.
- **Technical Leadership:** People Management, Client Communication, Product Management, Project Management

WORK EXPERIENCE

Senior Software Engineer (Applied AI) Sept 2025 - Present
Warner Bros. Discovery

- Led the design and development of a large-scale natural language video search and retrieval system across 1M+ hours of film and TV content, introducing architectural improvements that **increased retrieval performance by 15%**.
- Built a comprehensive **evaluation and benchmarking** framework to systematically compare embedding models, re-rankers, LLMs, prompting strategies, and hybrid retrieval approaches, enabling data-driven model selection.
- Established the tooling for observability and defined org-wide best practices for GenAI applications, improving reliability, evaluation rigor, and built team-wide harness for agentic coding.

Senior Software Engineer (Technical Lead) Apr 2025 - Sept 2025
reAlpha Tech Corp.

- Led the technical development of a home buying platform handling **5 million+ property data**. Built AI features that served natural language queries for home search, and automated internal real estate operations like scheduling tours and following up with leads. Technologies used: **(Python, Django, FastAPI, OpenAI, PostgreSQL, Elasticsearch)**
- Developed AI agents to automate information extraction and validation from loan documents (paystubs, W-2s, etc) **reducing processing time by 90%**.
- Redesigned a legacy NLP based conversational AI SaaS platform (AiChat) into an **LLM-based system** with multi-agent orchestration, Retrieval Augmented Generation (RAG) pipeline. Set up CI/CD using **Docker, Terraform and Github Actions** and managed deployments to **AWS ECS** as a multi-tenant microservice.

Co-founder & Chief Technology Officer Sep 2020 - Apr 2025
Naamche Inc (acquired by reAlpha Tech Corp.)

- Co-founded and scaled an AI software development studio from **3 to 40 engineers**; $0 to $500k ARR, built products for numerous clients, and led to a **successful acquisition** by reAlpha Tech Corp.
- Led the backend and platform engineering **(Node.js, Typescript, GraphQL, AWS)** of a property management software for a client (myAlphie) that was later sold to Crawford Hoying for $6 million.
- Engineered a real estate property evaluation engine that scored properties on their investment potential by using **predictive modeling** algorithms to analyze spatio-temporal and socio-economic data. Wrote the CI/CD pipelines and infrastructure as code (Terraform) to deploy the system to **AWS (Sagemaker, ECS and Lambdas)**.
- Developed an **information retrieval** tool for an agri-tech startup (Agrichat) with a RAG pipeline leveraging hybrid search (LLM embeddings, SPLADE) and re-ranking for better search results.
- Set up **observability and monitoring** (Grafana, Prometheus, Cloudwatch) to profile and optimize Django based fintech platform (EthanAI) and **reduced API response latency by 70%**.
- As the CTO, handled **end-to-end client communications**, requirements gathering, full-stack software development and progress reporting to ensure all customers are satisfied and business objectives are met.

Machine Learning Engineer Sep 2019 - Sep 2020
Fusemachines

- Developed an Optical Character Recognition system “FuseExtract” that digitized Nepali and English documents from their photos; tested successfully by a major bank and telecom company in Nepal.
- Leveraged data science techniques like predictive modeling, data mining, natural language processing and time series analysis to forecast the sales and revenue of video games for Take 2 Interactive.

EDUCATION & TRAINING

Bachelors in Computer Engineering Dec 2015 - Sep 2019
Institute of Engineering, Tribhuvan University Pulchowk, Nepal

- Graduated with **FIRST** position among all students in the batch with aggregate score of **85.3%**
- Recipient of **Academic Excellence Award** for the years: 2016, 2017 and 2019.

MicroMasters in Artificial Intelligence Apr 2018 - Dec 2020
edX, Columbia University

- Credentials: CSMM.101x: Artificial Intelligence (AI), CSMM.102x: Machine Learning

RESEARCH PUBLICATIONS

Dotel, S., et.al. (2020). **Disaster Assessment from Satellite Imagery by Analysing Topographical Features Using Deep Learning**. *IVSP20, 86-92.*